



Inspiring Excellence

CSE471: System Analysis and Design
Assignment on Functional Requirements
Proposed Project Title: AutoBD — Multi-Pillar Car Marketplace & Reconditioned Import Platform

Group No: ##, CSE471 Lab Section: ##, [Semester] [Year]		
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Project Overview

AutoBD is a comprehensive web platform unifying the four ways Bangladeshi buyers acquire cars — brand-new dealership purchases, peer-to-peer used car transactions, Japanese reconditioned vehicle imports via auction, and aftermarket modification sourcing — into a single trusted marketplace with a consistent, transparent cost calculation engine running across all four pillars.

The platform's core differentiator is its Reconditioned Auction section, which operates as a marketplace of competing licensed bidding organizations rather than a single in-house import team. Buyers select their preferred agent based on track record, ratings, fee structure, and turnaround time, then collaborate with that agent through a live bidding dashboard to win cars at Japanese auctions. This model — closer to Upwork than to SAT Japan or BE FORWARD — creates competitive pressure among agents on service quality and gives buyers genuine choice and accountability. Two Bangladesh-specific features further distinguish the platform: Container Pooling (automatically grouping buyers into shared shipping containers to cut per-car freight costs by up to 30%) and a BRTA Registration Paper Value Tracker (surfacing the currently opaque resale value of a car's remaining import-eligible registration life).

Tech Stack:

Language: TypeScript, JavaScript

Framework: Next.js

Styling: TailwindCSS

Database: PostgreSQL

ORM: Prisma

Deployment: Vercel

External APIs:

OpenAI API: For the AI-powered car recommendation engine and Dream Car Research Hub intelligence (Member A).

SSLCommerz / bKash API: For escrow payment processing and agent fee collection (Member D).

Google Maps API: For dealership/service center locator and live vessel position tracking (Members A & C).

Exchange Rate / Currency API (e.g., Open Exchange Rates): For live JPY-to-BDT conversion in the landed cost calculator and auction dashboard (Member D).

User Roles:

1. Buyer: The primary user — browses all four pillars, selects a bidding organization, participates in auctions, purchases used cars, orders modifications, and tracks shipments.

2. Bidding Organization: A licensed Japanese auction agent — applies for platform approval, manages their public profile, conducts live bidding on behalf of assigned buyers, and responds to ratings and disputes.

3. Admin: System super-user — approves bidding organization applications, manages duty rate tables and car database, monitors platform analytics, and adjudicates disputes.

Functional Requirements

SL	Common Workflows
1	Registration and Login System: Users can create an account by selecting their role: Buyer (individual purchasing a new, used, or reconditioned car), Bidding Organization (a licensed Japanese auction agent applying for platform partnership), or Administrator. Buyers register with name, email, phone, and password. Bidding Organizations complete a multi-step application form covering company name, license number, track record, and fee structure — their account remains inactive until approved by an admin. All users log in securely to access their role-specific dashboards.
2	Admin Role & Access Management: The administrator reviews and approves or rejects Bidding Organization applications, verifies submitted license details and track records, and can suspend any active organization account if complaints or rating thresholds are breached. The admin also manages the platform's car brand/model database, moderates used-car listings flagged by buyers, monitors platform-wide analytics (total bids placed, successful imports, active listings, revenue by pillar), and configures NBR duty rate tables used by the unified landed cost calculator.

Module 1

Member	Feature	Description
Member A	Brand New Car Listings & Dealer Inquiry	A structured browsing section for official brand-new car listings organised by manufacturer (Toyota, Honda, Mitsubishi, Suzuki, etc.). Each listing shows variant and trim options with BD-market pricing, verified manufacturer warranty terms (length, coverage, authorised service centers), and a specifications breakdown. Buyers select their preferred variant and submit a direct inquiry that is handed off to the corresponding dealership partner. The platform acts as a lead and booking layer — it does not hold inventory. Dealer partners are managed by the admin, and each listing links to a service center map (Google Maps API) showing the nearest authorised workshop.
Member B	Used Car P2P Marketplace & Seller Dashboard	A peer-to-peer marketplace where individual sellers can create car listings with photos, mileage, condition notes, asking price, and optional inspection report upload. Ownership document verification is prompted at listing creation — sellers upload relevant BRTA documents,

		which are reviewed by the platform's verification layer before the listing goes live. Buyers can browse, filter by make/model/year/price/location, and submit offers. Sellers manage all their active listings, offers received, and listing status from a dedicated seller dashboard. Listings that pass document verification receive a "Verified Ownership" badge.
Member C	Bidding Organization Marketplace & Profile Pages	The platform's signature differentiator. A browsable directory of all admin-approved licensed bidding organizations, each with a detailed profile page: company name, years in operation, number of successful imports, average turnaround time from auction win to BD delivery, customer star rating, fee structure (agent fee as fixed amount or percentage of bid), and verified license status. Buyers compare organizations side-by-side before selecting one for their reconditioned car search. This creates competitive pressure among agents on service quality, not just price — distinguishing the platform from single-vendor importers. Ratings are written by verified buyers who completed a transaction with that organization.
Member D	Unified Landed Cost Calculator	A consistent cost calculation engine available across all four platform pillars — new cars, used cars, reconditioned imports, and modification bundles. For reconditioned cars: bid price (in JPY) is converted to BDT at live exchange rates, then NBR import duty is applied based on engine CC and vehicle age (using the admin-maintained duty rate table), plus shipping estimate, agent fee, and port handling — giving a single total landed cost figure. The calculator updates in real time as the bid price changes during a live auction. For new and used cars, it shows total on-road price including registration and token tax. This is the single most-used tool across the platform and is intentionally kept consistent so users always trust the number they see.
Member A	Dream Car Research Hub	A top-of-funnel content and research section covering popular JDM and BD-market car models. Each model page contains: full specifications and variant comparison across generations, common reliability issues (sourced from owner community data), and a Total Cost of Ownership calculator specific to Bangladesh — covering registration tax, annual BRTA

		token tax, estimated fuel cost by engine size and BD fuel prices, insurance estimate, and typical parts availability and maintenance cost. Casual browsers not yet ready to buy are converted into platform leads through this section. Each model page links directly to relevant auction listings, new car pages, or the used car marketplace.
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Module 2

Member	Feature	Description
Member B	Reconditioned Car Auction Interface	The core auction experience for buyers who have selected a bidding organization. Buyers specify their target car using a structured form: make, model, year range, budget ceiling, maximum mileage, and must-have features, with an option to attach reference photos or links. The live bidding dashboard shows the current lot photo, auction sheet grade (A, B, C, D — with a translated condition breakdown), current bid price in JPY and live BDT equivalent, time remaining, and the number of active bidders. A real-time chat and call-log panel connects the buyer directly with their assigned bidder for live decisions (e.g., "go up to ¥750,000"). Buyers receive in-app and SMS alerts when a matching lot appears or when their bidder is about to place a bid.
Member C	Container Pooling — Group Shipping Coordination	A Bangladesh-specific feature addressing the high per-car shipping cost faced by individual buyers. After a buyer wins a lot at auction, the system checks for other buyers whose cars are at the same origin port (e.g., Yokohama) within a compatible shipping window and size class. Eligible buyers are shown a prompt: "3 open slots on a Yokohama→Chattogram container departing in 11 days — join to reduce your shipping cost by approximately 30%." Buyers who opt in are grouped into a shared container managed by the platform's logistics partner. The feature improves as the platform grows (more simultaneous imports = better pool matches), creating a natural network-effect moat that single-vendor importers cannot replicate.
Member D	Escrow Payment & Buyer Protection	A secure escrow-style payment flow for used car P2P transactions. When a buyer agrees to purchase a used car, funds are held in escrow via the payment gateway

	(SSLCommerz / bKash API)	(SSLCommerz or bKash) rather than transferred directly to the seller. The funds are released to the seller only after the buyer confirms receipt and satisfaction (or after a defined dispute window closes without a claim). For reconditioned auction purchases, the system collects the agent fee and shipping deposit via the same gateway and issues an itemised digital receipt. All payment records are stored per transaction and visible in both buyer and seller dashboards.
Member A	AI-Powered Car Recommendation Engine (OpenAI API)	A natural-language assistant that helps buyers who are unsure of what to look for. A buyer can describe their needs in plain text (e.g., "I want a fuel-efficient family car under 25 lakh taka, preferably Japanese, good for Dhaka traffic") and the system uses the OpenAI API to extract requirements, map them to available inventory across all three pillars (new, used, reconditioned), and return a ranked shortlist with reasoning for each recommendation. The assistant also flags trade-offs (e.g., "this model has lower fuel cost but higher parts cost in BD") using data from the Dream Car Research Hub. Buyers can refine suggestions through follow-up messages.
Member B	Modification Studio & Fitment Checker	A catalog browsing section for BRTA-legal cosmetic and performance modifications: wheels (Rays, BBS, Work, Enkei), body kits (Rocket Bunny, Modellista, Kuhl Racing), interior accessories, and lighting upgrades. Each product listing specifies compatibility by chassis code, bolt pattern, and offset. A fitment checker tool lets buyers input their specific car (or select from their garage) and filters the catalog to show only compatible parts — preventing incorrect orders. Modifications can be sourced through the platform's existing licensed bidding organization network (verified genuine parts), and buyers can bundle a modification order with their car purchase for combined shipping and pre-installation before delivery.

Module 3

Member	Feature	Description
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Member C	Shipment & Import Status Tracker (Google Maps API)	A real-time shipment tracking dashboard for buyers with an active import order. Stages tracked include: auction win confirmed → payment received → vehicle collected at Japanese port → vessel departed → vessel in transit → arrived at Chattogram port → customs clearance → ready for delivery. Each stage is timestamped and displayed on a visual progress timeline. A map view (Google Maps API) shows the vessel's last known position during transit. SMS and in-app notifications are sent automatically at each stage transition. Buyers with container-pooled shipments see all cars in their shared container on the same tracker.
Member D	BRTA Registration Paper Value Tracker	A Bangladesh-specific feature addressing a real, currently opaque market factor: a car's BRTA registration paper carries separate resale value tied to NBR import age-quota rules that shift with each national budget. For any used or reconditioned listing, this tool displays how many years of import-eligible registration life remain before the car ages out of common resale categories, along with a historical view of how NBR's import age-limit policy has changed across past budgets. This converts a word-of-mouth, informal factor into a transparent, data-backed display on every relevant listing — a genuine information advantage for buyers that no existing platform currently offers.
Member A	Accident History & Inspection Verification Report	For used car listings, sellers can upload auction sheets, previous inspection reports, and ownership transfer documents. The platform's verification layer cross-references the chassis number (VIN) against a partner database of accident records and flags any history found. Verified listings display an "Accident History: None Found" or "1 Incident Recorded" badge. Buyers can optionally request a third-party physical inspection through a partner garage network, booked and paid directly through the platform. The inspection report is uploaded to the listing and permanently visible to all subsequent buyers if the car is relisted — creating a persistent trust record tied to the chassis number.
Member B	Bidding Organization Rating & Dispute System	After a reconditioned car import is completed and delivered, the buyer is prompted to rate their bidding organization across four dimensions: communication quality, accuracy of auction sheet grading, delivery

		timeline adherence, and overall value. Ratings contribute to the organization's public profile score. If a buyer disputes a transaction (e.g., car condition significantly differs from described grade), they can raise a formal dispute through the platform. The admin reviews the dispute with access to the full chat and bid log. Repeat disputes against an organization above a configurable threshold trigger an automatic suspension pending admin review, maintaining marketplace quality without requiring manual monitoring of every transaction.
Member C & D	Admin — Platform Analytics & Revenue Dashboard	A system-wide admin view displaying key platform metrics across all four pillars: total active listings by category, total bids placed and auction win rate, successful imports completed, average landed cost by vehicle class, container pooling match rate, escrow transaction volume, and revenue breakdown by source (dealership referral commissions, listing fees, agent placement commissions, modification sourcing margin). Charts display trends over configurable date ranges. The admin can also manage NBR duty rate tables (updated each budget cycle), adjust container pooling parameters, and view all pending Bidding Organization applications and active disputes from a single panel.